Extra Dessert #1: Specialized Propagation

EME, meteor scatter, tropospheric ducting, auroral propagation, and the physics of electromagnetic waves.

Group E3A · 14 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

Electromagnetic Wave Basics

Question	Answer	Q
Direction of EM wave travel	**At a right angle to the electric and magnetic fields.**	E3A04
Orientation of E and B fields	**At right angles** to each other.	E3A05
What determines wave speed in a medium?	**The index of refraction** of the medium.	E3A10
Circular polarization	**Waves with rotating electric and magnetic fields.** (Common in satellite work.)	E3A14

Earth-Moon-Earth (EME / "Moonbounce")

Topic	Answer	Q
Maximum separation between EME stations	About **12,000 miles**, as long as the moon is visible from both stations.	E3A01
Libration fading	**Fluttery, irregular fading** caused by the moon's surface wobble and uneven reflection.	E3A02
Least path loss	When the moon is at **perigee** (closest to Earth).	E3A03

Meteor Scatter

Topic	Answer	Q
Where meteor ionization forms	The **E region** of the ionosphere.	E3A08
Best frequency range	**28 MHz to 148 MHz** (10 m through 2 m).	E3A09

Tropospheric Ducting and Auroral Propagation

Topic	Answer	Q
Where atmospheric ducts form (microwave)	Over **large bodies of water**.	E3A07
Typical tropospheric duct range	**100 to 300 miles**.	E3A11
What causes auroral propagation	**Severe geomagnetic storms**.	E3A12
Best mode for auroral propagation	**CW** – auroral signals are heavily distorted, so the simplest mode wins.	E3A13

Daily MUF Variation

Situation	Action	Q
MUF drops as darkness arrives during a long-distance contact	**Switch to a lower-frequency HF band.** As ionization drops, only lower frequencies still refract back to Earth.	E3A06

Dessert 1 Question Bank (E3A)

E3A01. What is the approximate maximum separation measured along the surface of the Earth between two stations communicating by EME?
A. 2,000 miles, if the moon is at perigee.
B. 2,000 miles, if the moon is at apogee.
C. 5,000 miles, if the moon is at perigee.
D. 12,000 miles, if the moon is “visible” by both stations.

E3A02. What characterizes libration fading of an EME signal?
A. A slow change in the pitch of the CW signal.
B. A fluttery, irregular fading.
C. A gradual loss of signal as the sun rises.
D. The returning echo is several hertz lower in frequency than the transmitted signal.

E3A03. When scheduling EME contacts, which of these conditions will generally result in the least path loss?
A. When the Moon is at perigee.
B. When the Moon is full.
C. When the Moon is at apogee.
D. When the MUF is above 30 MHz.

E3A04. In what direction does an electromagnetic wave travel?
A. It depends on the phase angle of the magnetic field.
B. It travels parallel to the electric and magnetic fields.
C. It depends on the phase angle of the electric field.
D. It travels at a right angle to the electric and magnetic fields.

E3A05. How are the component fields of an electromagnetic wave oriented?
A. They are parallel.
B. They are tangential.
C. They are at right angles.
D. They are 90 degrees out of phase.

E3A06. What should be done to continue a long-distance contact when the MUF for that path decreases due to darkness?
A. Switch to a higher frequency HF band.
B. Switch to a lower frequency HF band.
C. Change to an antenna with a higher takeoff angle.
D. Change to an antenna with greater beam width.

E3A07. Atmospheric ducts capable of propagating microwave signals often form over what geographic feature?
A. Mountain ranges.
B. Stratocumulus clouds.
C. Large bodies of water.
D. Nimbus clouds.

E3A08. When a meteor strikes the Earth’s atmosphere, a linear ionized region is formed at what region of the ionosphere?
A. The E region.
B. The F1 region.
C. The F2 region.
D. The D region.

E3A09. Which of the following frequency ranges is most suited for meteor-scatter communications?
A. 1.8 MHz - 1.9 MHz.
B. 10 MHz - 14 MHz.
C. 28 MHz - 148 MHz.
D. 220 MHz - 450 MHz.

E3A10. What determines the speed of electromagnetic waves through a medium?
A. Resistance and reactance.
B. Evanescence.
C. Birefringence.
D. The index of refraction.

E3A11. What is a typical range for tropospheric duct propagation of microwave signals?
A. 10 miles to 50 miles.
B. 100 miles to 300 miles.
C. 1,200 miles.
D. 2,500 miles.

E3A12. What is most likely to result in auroral propagation?
A. Meteor showers.
B. Quiet geomagnetic conditions.
C. Severe geomagnetic storms.
D. Extreme low-pressure areas in polar regions.

E3A13. Which of these emission modes is best for auroral propagation?
A. CW.
B. SSB.
C. FM.
D. RTTY.

E3A14. What are circularly polarized electromagnetic waves?
A. Waves with an electric field bent into a circular shape.
B. Waves with rotating electric and magnetic fields.
C. Waves that circle Earth.
D. Waves produced by a loop antenna.

Answer Key

E3A01: D	E3A02: B	E3A03: A	E3A04: D
E3A05: C	E3A06: B	E3A07: C	E3A08: A
E3A09: C	E3A10: D	E3A11: B	E3A12: C
E3A13: A	E3A14: B		